Fiscal Year:	FY 2012	**Task Last Updated:**	FY 07/02/2012
PI Name:	Kronenberg, Amy Sc.D.		
Project Title:	Comparative Analysis of Charged Particle-Induced Autosomal Mutations in Murine Tissues and Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-8099	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	04/01/2007	**End Date:**	03/31/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	4
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Project extended until 3/31/2012 per C. Guidry/JSC (Ed., 2/14/2011)		
Key Personnel Changes/Previous PI:	No changes		
COI Name (Institution):	Turker, Mitchell (Oregon Health & Science University) Grygoryev, Dmytro (Oregon Health & Science University) Dan, Cristian (Oregon Health & Science University) Gauny, Stacey (Lawrence Berkeley National Laboratory) Grossi, Gianfranco (Universita Federico II) Eckelmann, Bradley (Oregon Health & Science University)		
Grant/Contract No.:	NNJ07HC72I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Mutations are required in the transition from normal cells to cancer cells. Solid cancers in people arise primarily in epithelial cells. This proposal addresses the carcinogenic risk of exposure to energetic protons, the most abundant ion in galactic cosmic radiation (GCR) and solar particle events. The toxic effects of these exposures will also be measured. The studies outlined quantify mutation induction at a model, autosomal genetic locus (Aprt) located on mouse chromosome 8. C57BL/6 x DBA2 F1 Aprt heterozygous mice or kidney epithelial cells from this mouse strain will be irradiated to determine dose responses for cell killing and Aprt mutation in kidney epithelial cells exposed in the mice, or in kidney epithelial cells irradiated in culture. This will allow assessment of the role of the tissue microenvironment on modulation of mutation risk and on persistence of radiation-induced mutants in the kidney. Molecular analysis will determine the mechanisms leading to proton-induced Aprt mutants in exposed kidneys or in kidney cells irradiated in culture. It is predicted that proton exposure will result in mutants that lose the Aprt gene and other genes that are located in nearby regions of mouse chromosome 8. Studies are proposed to test if protracted exposure lessens the frequency of proton-induced mutants while also lowering the risk of killing kidney epithelial cells. The toxic and mutagenic effects of sequential exposure to protons and iron ions will be assessed (with iron ions serving as a representative heavy ion in the GCR) to test whether the risks are additive. Results from these studies will help reduce uncertainties in estimating the risk of cancer for astronauts exposed to protons alone or in mixed radiation fields. They will also provide detailed knowledge of the types of mutations that are created by these exposures that can contribute to the development of cancer. New studies are included to address the links between centrosome abnormalities, aneuploidy and mutagenesis as a function of dose, LET, and tissue microenvironment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will have a direct impact on our understanding of proton-induced toxicity and mutations in a mammalian epithelial tissue. This has relevance to cancer therapy involving proton exposures that not only target malignant tissue, but which also involve limited exposure of normal tissues that can develop late complications as a result of proton exposure. In addition, our in vivo/in vitro approach also illuminates the importance of the normal tissue environment on the modulation of radiation sequelae.
Task Progress:	A new focus of work in the final year of funding was the establishment of the dose response for cell killing for 240 MeV/amu Si ions, to begin to address portions of the supplemental aims. Experiments were performed in triplicate with kidney epithelial cells irradiated in vitro in the Fall 2011 NASA Space Radiation Laboratory (NSRL) run. The results indicate substantial cell killing, with Si ions nearly equal to Fe ions in their potency when the data are considered in terms of absorbed dose. Si ions were much more toxic to kidney epithelial cells in culture (per unit dose) than 1 GeV protons. Studies were also initiated to develop populations of cells in which to measure centrosome abnormalities and aneuploidy at the earliest times post-irradiation also in order to assess portions of the supplemental aims. Replicate cultures were exposed to graded doses of Fe ions and processed for future use. In addition, clonal isolates were developed of non-selected and Aprt mutants after exposure to low to moderate doses of Fe ions and were processed for future use. These studies were performed during the Spring and Fall 2011 NSRL runs. Similar studies were initiated in the Fall 2011 NSRL run for kidney clones derived from in situ exposures to graded doses of Fe ions. We have continued with the molecular analysis of changes in Aprt mutant and non-mutant clones following exposures to lower doses of Fe ions. Characteristic Fe ion-induced alterations initially identified in mutants arising after a high dose exposure can be readily detected at lower doses. Genome-wide analyses indicate that these characteristic events may involve other non-selected chromosomes and that they can also be found in subsets of Fe ion-induced mutants arising after lower dose exposures.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Grygoryev D, Dan C, Turker M. "Charged particle-induced autosomal mutagenesis in normal epithelium exposed to energetic protons or Fe ions." International Symposium for Radiation Research and Medical Physics, Shanghai, China, May 30-June 2, 2011. International Symposium for Radiation Research and Medical Physics, Shanghai, China, May 30-June 2, 2011. , Jun-2011
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Grygoryev D, Dan C, Turker M. "Charged particle induced autosomal mutagenesis in normal epithelium exposed to energetic protons or Fe ions." 14th International Congress of Radiation Research, Warsaw, Poland, August 28-September 1, 2011. 14th International Congress of Radiation Research, Warsaw, Poland, August 28-September 1, 2011. , Aug-2011
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Grygoryev D, Dan C, Turker M. "Charged particle-induced autosomal mutagenesis in normal epithelium exposed to energetic protons or Fe ions." 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7011.pdf , Sep-2011
Abstracts for Journals and Proceedings	Turker M, Dan C, Grygoryev D, Gauny S, Kwoh E, Kronenberg A. "Viable cells with extensive genomic change from mouse epithelium exposed to charged particles." 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7026.pdf , Sep-2011
Articles in Peer-reviewed Journals	Kronenberg A, Cucinotta FA. "Space radiation protection issues." Health Physics, in press, 2012. Expected publication November 2012. , Jul-2012